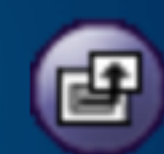


1. A 19-year-old woman comes for a routine follow-up examination for acne vulgaris. She is sexually active and does not use contraception. She has had chronic migraines over the past 3 years. She has a history of early signs of cataracts. There is a family history of hypercholesterolemia and inflammatory bowel disease. Examination shows active acne vulgaris on her face. She says that the various treatments she has received have failed to adequately control her acne, and she asks for a prescription for isotretinoin, a medication that has worked well for many of her friends. Isotretinoin therapy is contraindicated in this patient because of which of the following historic findings?
- ☐ A) Chronic migraines
 - ☐ B) Early cataracts
 - ☐ C) Family history of hypercholesterolemia
 - ☐ D) Family history of inflammatory bowel disease
 - ☐ E) Sexual activity without contraception



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

2. A 73-year-old man with hypertension is brought to the emergency department by paramedics 1 hour after the onset of moderate chest pain and difficulty breathing. Aspirin was administered en route to the hospital. He takes no other medications. On arrival, he is diaphoretic. His temperature is 37.2°C (98.9°F), pulse is 185/min and thready, respirations are 30/min, and blood pressure is 85/45 mm Hg. Pulse oximetry on 100% oxygen by face mask shows an oxygen saturation of 92%. Breath sounds are heard bilaterally. Cardiac examination shows a regular rhythm. A rhythm strip shows a wide complex tachycardia. Which of the following is the most appropriate initial step in management?
- ☐ A) Administration of esmolol
 - ☐ B) Administration of nitroglycerin by intravenous drip
 - ☐ C) Intravenous administration of morphine
 - ☐ D) Endotracheal intubation
 - ☐ E) Synchronized cardioversion



3. A 55-year-old woman comes to the emergency department 30 minutes after the sudden onset of ripping substernal chest pain. She has hypertension treated with hydrochlorothiazide. Her blood pressure is 220/128 mm Hg in both arms. Carotid, radial, femoral, and pedal pulses are within normal limits. An x-ray of the chest shows a widened aortic root and dilated ascending aorta. Which of the following is the most likely cause?
- ☐ A) Aortic dissection due to Takayasu arteritis
 - ☐ B) Degeneration of the aortic wall from Marfan syndrome
 - ☐ C) Medial degeneration of the aorta
 - ☐ D) Rupture of a sinus of Valsalva aneurysm
 - ☐ E) Traumatic disruption of the aorta from a fall



4. A 27-year-old African American woman comes to the physician because of a 1-week history of a tight sensation in her chest and a nonproductive cough. The cough occurs most frequently at night after she goes to bed. She says she received breathing treatments in the emergency department for acute shortness of breath twice during the past 3 months and was told to stop smoking. She has a long-standing history of frequent upper respiratory tract infections. She currently takes no medications. She has smoked one pack of cigarettes daily for 10 years. Vital signs are within normal limits. End-expiratory wheezes are heard bilaterally. The remainder of the examination shows no abnormalities. A chest x-ray shows no abnormalities. Which of the following is the most likely underlying cause of this patient's symptoms?
- ☐ A) Activation of mast cells
 - ☐ B) Embolism of intravascular thrombi
 - ☐ C) Loss of elastin in the lung matrix
 - ☐ D) Necrotizing pulmonary vasculitis
 - ☐ E) Noncaseating granulomatous inflammation



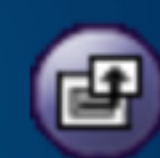
5. Immediately after an episode of massive hematemesis, a 50-year-old man feels faint, cold, and sweaty. His pulse is 130/min, and blood pressure is 80/40 mm Hg; hematocrit is 40%. Which of the following is the most appropriate next step in patient care?
- ☐ A) Rapid infusion of 5% dextrose in water
 - ☐ B) Rapid infusion of 0.9% isotonic saline
 - ☐ C) Insertion of a pulmonary artery catheter via the subclavian vein
 - ☐ D) Gastric lavage with ice-cold saline until bleeding stops
 - ☐ E) Emergency esophagogastroduodenoscopy



Previous



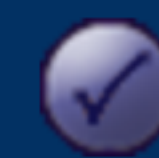
Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

6. A 72-year-old woman is brought to the emergency department because of a 1-month history of progressive shortness of breath and fatigue. She has had no chest pain. There is no history of heart or pulmonary disease. She is in moderate respiratory distress. Her temperature is 37°C (98.6°F), pulse is 100/min, respirations are 22/min, and blood pressure is 135/85 mm Hg. Examination shows jugular venous distention. Crackles are heard halfway up both lungs. A grade 2/6, systolic ejection murmur is heard along the left sternal border. There is an S₃. She has 2+ edema of the lower extremities. Rectal examination shows dark stool; test for occult blood is positive. Laboratory studies show:

Hemoglobin	5 g/dL
Leukocyte count	9000/mm ³
Serum	
Na ⁺	140 mEq/L
K ⁺	4 mEq/L
Cl ⁻	105 mEq/L
HCO ₃ ⁻	25 mEq/L
Urea nitrogen	28 mg/dL
Glucose	120 mg/dL
Creatinine	1.2 mg/dL
Urine protein	1+

An x-ray of the chest shows a mildly enlarged cardiac silhouette with pulmonary vascular congestion. An ECG shows nonspecific ST-segment changes. Echocardiography shows trace mitral regurgitation and an ejection fraction of 70%; there are no wall motion abnormalities. Which of the following is the most likely cause of these findings?

- ☐ A) High-output heart failure
- ☐ B) Left ventricular diastolic dysfunction
- ☐ C) Left ventricular systolic dysfunction
- ☐ D) Nephrotic syndrome
- ☐ E) Valvular heart disease

7. A 47-year-old man comes to the physician because of a 4-day history of abdominal bloating, increasing flatulence, and diarrhea 6 to 12 times daily despite loperamide therapy. The stool is watery, nonbloody, and foul smelling; he says it is usually brown and occasionally green. He has not had fever. He recently traveled to the Middle East. His temperature is 37°C(98.6°F), pulse is 88/min, respirations are 18/min, and blood pressure is 120/76 mm Hg. Abdominal examination shows slight distention and mild tenderness with no rebound or guarding; bowel sounds are normal. Rectal examination shows no abnormalities. Test of the stool for occult blood is negative. Which of the following is the most likely causal organism?

- | | |
|--|---|
| ☐ A) <i>Campylobacter jejuni</i> | ☐ F) Nontyphoidal <i>Salmonella</i> species |
| ☐ B) <i>Clostridium difficile</i> | ☐ G) Norwalk virus |
| ☐ C) <i>Entamoeba coli</i> | ☐ H) Rotavirus |
| ☐ D) <i>Entamoeba histolytica</i> | ☐ I) <i>Shigella</i> species |
| ☐ E) Enterotoxigenic <i>Escherichia coli</i> | ☐ J) <i>Yersinia enterocolitica</i> |



8. A 32-year-old woman has had polyuria, blurred vision, and rapid breathing for 1 day. Her pulse is 110/min, respirations are 26/min, and blood pressure is 100/50 mm Hg. Examination is otherwise within normal limits. Serum studies show:

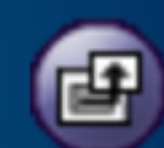
Na ⁺	139 mEq/L
Cl ⁻	99 mEq/L
K ⁺	4.7 mEq/L
HCO ₃ ⁻	18 mEq/L
Glucose	783 mg/dL

Which of the following is the most likely cause of her acidosis?

- ☐ A) Accumulation of β-hydroxybutyric acid
- ☐ B) Accumulation of lactate
- ☐ C) Decreased renal production of ammonium
- ☐ D) Loss of bicarbonate through the kidneys
- ☐ E) Movement of potassium into the cells

9. A 27-year-old man comes to the physician because of a 3-hour history of muscle swelling and tenderness of the left thigh following minimal trauma. He has severe hemophilia A and has received factor VIII replacement therapy for bleeding episodes since childhood. Over recent months, his bleeding has been poorly controlled; increasing doses of factor VIII have been required to stop bleeding episodes. His platelet count is $125,000/\text{mm}^3$, and activated partial thromboplastin time is greater than 120 sec. Which of the following is the most appropriate next step in diagnosis of this patient's increased bleeding?

- ☐ A) Measurement of bleeding time
- ☐ B) Measurement of factor IX concentration
- ☐ C) Measurement of thrombin time to detect heparin
- ☐ D) Platelet function studies
- ☐ E) Test for factor VIII inhibitor



10. A 23-year-old African American man comes to the physician because of a 2-month history of increasing pain and swelling above the right knee. There is no history of trauma or recent fever. He had retinoblastoma in infancy that was treated with enucleation. His temperature is 37°C (98.6°F). Examination shows moderate swelling of the distal femur. There is mild tenderness to palpation with moderate discomfort on weight bearing. His serum alkaline phosphatase activity is 1350 U/L. An x-ray of the right femur shows a lytic lesion with periosteal new bone at the margins. Which of the following is the most likely diagnosis?

- ☐ A) Hyperparathyroidism
- ☐ B) Metastatic malignancy
- ☐ C) Multiple myeloma
- ☐ D) Osteitis deformans (Paget disease)
- ☐ E) Osteoid osteoma
- ☐ F) Osteomyelitis
- ☐ G) Osteoporosis
- ☐ H) Osteosarcoma
- ☐ I) Sickle cell disease



11. A previously healthy 21-year-old man comes to the emergency department 2 weeks after the onset of mild cough and runny nose. During the past 3 days, the cough has worsened with episodes so intense that he has vomited. The cough is productive of thick green sputum. He feels well between coughing episodes. He has not had fever, chest pain, or shortness of breath. He takes no medications. He is a college student. His temperature is 37.3°C (99.2°F), pulse is 70/min, respirations are 14/min, and blood pressure is 115/65 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. Pulmonary examination shows no abnormalities. Which of the following is the most likely diagnosis?
- ☐ A) Influenza
 - ☐ B) Legionnaires disease
 - ☐ C) Pertussis
 - ☐ D) Psittacosis
 - ☐ E) Q fever



12. A 27-year-old woman who is HIV positive comes to the physician because of a 1-day history of fever, shortness of breath, right-sided chest pain, and cough productive of yellow-brown sputum. She had previously been feeling well. She is compliant with her antiretroviral medication regimen and follow-up visits. One month ago, her CD4+ T-lymphocyte count was 400/mm³ (Normal ≥500), and plasma HIV viral load was undetectable. She has never used illicit drugs. Her temperature is 39.8°C (103.6°F), pulse is 120/min, respirations are 28/min, and blood pressure is 90/40 mm Hg. Bronchial breath sounds and egophony are heard over the lower lobe of the right lung posteriorly; there is dullness to percussion. Pulse oximetry on room air shows an oxygen saturation of 93%. An x-ray of the chest shows a right lower lobe consolidation. A Gram stain of sputum shows many segmented neutrophils and numerous gram-positive diplococci. Cultures of the sputum and blood are pending. In addition to supplemental oxygen and intravenous fluids, which of the following is the most appropriate pharmacotherapy?

- ☐ A) Ceftriaxone
- ☐ B) Gentamicin
- ☐ C) Isoniazid, rifampin, and pyrazinamide
- ☐ D) Penicillin
- ☐ E) Prednisone
- ☐ F) Trimethoprim-sulfamethoxazole



13. A 35-year-old woman is brought to the emergency department after collapsing at a restaurant. Her pulse is 120/min, respirations are 30/min, and blood pressure is 90/60 mm Hg. Her breathing is labored, but she can speak. She has diffuse bronchospasm, and she coughs continuously. Which of the following is the most likely diagnosis?

- ☐ A) Anaphylactic shock
- ☐ B) Aspiration
- ☐ C) Drug overdose
- ☐ D) Pulmonary embolus
- ☐ E) Spontaneous pneumothorax



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

14. A 42-year-old woman comes to the physician for a follow-up examination. She has a 5-year history of mild shortness of breath on exertion and cough productive of copious yellow-to-whitish sputum. She has not had fever or chills. She has reactive airways disease treated with daily salmeterol-fluticasone and albuterol and seasonal allergies treated with fexofenadine. She has a chronic cough that is not controlled by her medications. She has a 20-year history of multiple respiratory tract infections, including three episodes of pneumonia during the past 15 years and two episodes of sinusitis during the past 5 years. She usually requires antibiotic therapy for bronchitis three to four times yearly. She does not smoke. Her temperature is 36.9°C (98.4°F), pulse is 72/min, respirations are 20/min, and blood pressure is 132/78 mm Hg. Examination shows mild lymphadenopathy. There is mild thrush. Coarse breath sounds are heard bilaterally. Cardiac examination shows no abnormalities except for a slightly increased S₂. There is no clubbing or edema. In addition to pulmonary function testing and antifungal therapy, which of the following is the most appropriate next step in diagnosis?

- ☐ A) Measurement of CD4+ T-lymphocyte count
- ☐ B) Measurement of serum complement concentrations
- ☐ C) Measurement of serum immunoglobulin concentrations
- ☐ D) Sputum culture
- ☐ E) Ventilation-perfusion lung scans



15. A 42-year-old woman comes to the physician because of a 3-day history of dizziness and progressive difficulty with balance. She has been supporting herself against walls when she walks. She has not had numbness, weakness, or palpitations. She has had nausea and a mild decrease in appetite but is able to drink fluids. There is no history of head trauma. She has mitral valve prolapse. She takes no medications. Her sister has multiple sclerosis. Vital signs are within normal limits. The optic discs appear normal, and ocular movements are full. Tympanic membranes are dull bilaterally. The light reflex is distorted. There are no exudates on the pharynx. A grade 2/6 systolic murmur is heard best at the lower left sternal border. Muscle strength and deep tendon reflexes are normal. Her gait is unsteady. Which of the following is the most likely diagnosis?

- ☐ A) Acoustic neuroma (vestibular schwannoma)
- ☐ B) Cerebral infarction
- ☐ C) Dehydration
- ☐ D) Endocarditis
- ☐ E) Labyrinthitis
- ☐ F) Multiple sclerosis



16. A 57-year-old man comes to the physician with his wife because of a headache for 3 days. He has had a 6.8-kg (15-lb) weight gain over the past 6 months. His dentist has told him that the space between his teeth is increasing. His wife says that he has started snoring and his voice has deepened over the past 6 months. He is 178 cm (5 ft 10 in) tall and now weighs 91 kg (200 lb); BMI is 29 kg/m². Examination shows frontal bossing and large feet. Measurement of which of the following is most likely to confirm the diagnosis?

- ☐ A) Hemoglobin A_{1c}
- ☐ B) Serum C-peptide concentration
- ☐ C) Serum growth hormone-releasing hormone concentration
- ☐ D) Serum insulin-like growth factor 1
- ☐ E) Serum somatostatin concentration



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

17. A previously healthy 72-year-old man who is a retired shipyard worker comes to the physician because of a 2-month history of nonproductive cough and shortness of breath. He does not smoke. Breath sounds are decreased, and there is dullness to percussion at the left lung base. An x-ray of the chest shows calcified pleural plaques and a left pleural effusion. A thoracentesis is performed, and analysis of the pleural fluid shows a glucose concentration of 4 mg/dL. The most likely underlying cause of these findings is exposure to which of the following?

- ☐ A) Asbestos
- ☐ B) Graphite
- ☐ C) Isocyanates
- ☐ D) *Mycobacterium kansasii*
- ☐ E) *Mycoplasma pneumoniae*
- ☐ F) *Nocardia asteroides*
- ☐ G) Radon
- ☐ H) Silicon



18. A 57-year-old man comes to the physician because of shortness of breath and wheezing for 2 days. He also has had swelling of the lower legs since he returned from a 200-mile car trip 3 days ago. He has a 20-year history of poorly controlled hypertension currently treated with hydrochlorothiazide. His temperature is 37°C (98.6°F), pulse is 100/min and regular, respirations are 24/min, and blood pressure is 170/100 mm Hg. Examination shows jugular venous distention. Crackles and wheezes are heard in the lower lung fields bilaterally. A prominent S₃ gallop is heard at the apex. There is 4+ ankle edema. An x-ray of the chest shows cardiomegaly and cephalization of the pulmonary vessels. Which of the following is the most likely diagnosis?
- ☐ A) Aspiration pneumonia
 - ☐ B) Bronchiectasis
 - ☐ C) Chronic bronchitis
 - ☐ D) Hypersensitivity pneumonitis
 - ☐ E) Pulmonary edema
 - ☐ F) Pulmonary embolism



19. A previously healthy 22-year-old woman comes to the physician because of a 2-day history of fever, chills, and left flank pain. She also has had nausea and vomited four times during this period. Her temperature is 38.9°C (102°F), pulse is 110/min, and blood pressure is 90/60 mm Hg. The abdomen is soft with tenderness to percussion over the left flank. The remainder of the examination shows no abnormalities. Laboratory studies show:

Hematocrit	39%
Leukocyte count	22,000/mm ³
Urine	
RBC	10–20/hpf
WBC	20–50/hpf

Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Oral amoxicillin
- ☐ B) Oral azithromycin
- ☐ C) Intravenous amoxicillin
- ☐ D) Intravenous ceftriaxone
- ☐ E) Intravenous metronidazole

20. A previously healthy 27-year-old woman comes to the physician because of swelling of her face and legs for 2 weeks. As the day progresses, her facial swelling resolves, but her legs become increasingly swollen. Her pulse is 64/min and regular, and blood pressure is 110/70 mm Hg. Cardiopulmonary examination shows no abnormalities. There is 1+ presacral edema and 2+ edema of the lower extremities. Her prothrombin time is 12 seconds (INR=1.1), and serum albumin concentration is 2.3 g/dL. Which of the following is the most likely cause of this patient's edema?

- | | |
|---|---|
| ☐ A) Cellulitis | ☐ G) Malnutrition |
| ☐ B) Cirrhosis | ☐ H) Nephrotic syndrome |
| ☐ C) Congestive heart failure | ☐ I) Postphlebitic syndrome |
| ☐ D) Deep venous thrombosis | ☐ J) Pulmonary hypertension |
| ☐ E) Lymphangitis | ☐ K) Stasis dermatitis |
| ☐ F) Lymphedema | ☐ L) Varicose veins |



21. A 30-year-old woman with systemic lupus erythematosus is brought to the emergency department 2 hours after the onset of severe substernal chest pain that increases with respiration. She appears distressed and is bent over at the waist to decrease the pain. She has no history of cough, chills, or fever. Her temperature is 37.6°C (99.7°F), pulse is 100/min and regular, respirations are 20/min, and blood pressure is 120/80 mm Hg. A short, harsh diastolic sound is heard on auscultation of the chest; breath sounds are normal. Which of the following is the most likely diagnosis?

- ☐ A) Acute myocardial infarction
- ☐ B) Aortic dissection
- ☐ C) Endocarditis
- ☐ D) Pericarditis
- ☐ E) Pulmonary embolus



22. A 74-year-old woman has a myocardial infarction and is admitted to the intensive care unit. Her blood pressure has decreased from 148/74 mm Hg to 80/62 mm Hg. She is confused and has cool clammy skin. Arterial blood gas analysis is most likely to show which of the following?

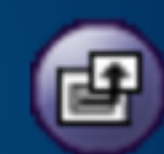
- ☐ A) Hypoxemia with normal pH
- ☐ B) Primary metabolic acidosis
- ☐ C) Primary metabolic alkalosis
- ☐ D) Primary respiratory acidosis
- ☐ E) Primary respiratory alkalosis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

23. A 65-year-old man has a positive test of the stool for occult blood during a routine examination. He has no personal or family history of gastrointestinal disorders. Which of the following is the most appropriate next step in evaluation?

- ☐ A) Repeat test of the stool for occult blood in 6 weeks
- ☐ B) Measurement of serum carcinoembryonic antigen (CEA) concentration
- ☐ C) CT scan of the abdomen
- ☐ D) Flexible sigmoidoscopy
- ☐ E) Colonoscopy



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

24. An 82-year-old man has been mechanically ventilated in the hospital for the past 48 hours because of an exacerbation of chronic obstructive pulmonary disease. He has passed no urine since removal of a urinary catheter 12 hours ago. Current medications include intravenous methylprednisolone and albuterol, ipratropium, and fluticasone by inhalation. Which of the following is the most likely diagnosis?

- ☐ A) Acute interstitial nephritis
- ☐ B) Acute urethral obstruction
- ☐ C) Contrast-induced nephropathy
- ☐ D) End-stage renal disease
- ☐ E) Hepatorenal syndrome
- ☐ F) Ischemic acute tubular necrosis
- ☐ G) Myoglobinuric acute renal failure (rhabdomyolysis)
- ☐ H) Prerenal azotemia



25. A 52-year-old man with a 5-year history of essential hypertension comes to the physician for a follow-up examination. He has read about a clinical trial for a new medication, drug X, which has shown promise in controlling difficult cases of hypertension, and inquires about participation. His hypertension is moderately controlled with a thiazide diuretic, β -adrenergic blocking agent, and calcium-channel blocking agent. His temperature is 37°C (98.6°F), pulse is 80/min, respirations are 12/min, and blood pressure is 150/95 mm Hg. Examination shows no other abnormalities. Which of the following is the most appropriate study design?
- ☐ A) Prospective observational cohort
 - ☐ B) Retrospective observational cohort
 - ☐ C) Randomized controlled trial of drug X versus a placebo
 - ☐ D) Randomized controlled trial of drug X versus standard therapy



26. A 27-year-old man with moderate mental retardation is brought to the physician by his caretaker because of increasingly frequent episodes of intermittent vomiting over the past 3 weeks. The vomiting typically occurs within an hour after eating and usually contains partially digested food; no blood is noted in the vomitus. The patient's appetite seems unaffected, but he stops eating before he completes the meal. He has had a 3.6-kg (8-lb) weight loss since his last visit 6 months ago. He takes phenytoin for a seizure disorder but is otherwise healthy. His vital signs are within normal limits. His speech is not intelligible. Examination shows a 3 x 4-cm area of alopecia in the right occipitoparietal area of the scalp. Bowel sounds are present. The abdominal musculature voluntarily contracts with palpation; no organomegaly or masses are palpated. There are minimal contractures in both upper and lower extremities and increased muscle tone. Which of the following is the most likely diagnosis?

- | | |
|---|--|
| ☐ A) Achalasia | ☐ G) Food poisoning |
| ☐ B) Brain tumor | ☐ H) Gastric bezoar |
| ☐ C) Bulimia nervosa | ☐ I) Gastric carcinoma |
| ☐ D) Cholecystitis | ☐ J) Pyloric channel ulcer |
| ☐ E) Diabetic gastroparesis | ☐ K) Small-bowel obstruction |
| ☐ F) Drug toxicity | ☐ L) Uremia |



27. A previously healthy 27-year-old woman comes to the emergency department because of a 2-day history of fever and right knee pain. She has not had cough, abdominal pain, or pain with urination. There is no history of intravenous drug use. She appears uncomfortable. Her temperature is 39°C (102.2°F), pulse is 110/min, respirations are 12/min, and blood pressure is 110/70 mm Hg. Examination shows a few raised erythematous papules on both palms. The right knee is warm, erythematous, and tender to palpation. There is a moderate-sized effusion. Passive flexion of the knee is limited to 30 degrees because of severe pain. Analysis of synovial fluid aspirated from the right knee shows a leukocyte count of 45,000/mm³ (80% segmented neutrophils and 20% lymphocytes). A Gram stain of the synovial fluid and microscopic examination for crystals are negative. Which of the following is the most likely underlying cause of these findings?
- ☐ A) A contiguous soft tissue infection of the joint by *Staphylococcus aureus*
 - ☐ B) Immunologic-mediated arthritis secondary to *Borrelia burgdorferi*
 - ☐ C) Immunologic-mediated arthritis secondary to group A streptococcus
 - ☐ D) Infection of the joint secondary to disseminated *Borrelia burgdorferi*
 - ☐ E) Infection of the joint secondary to disseminated *Neisseria gonorrhoeae*



The response options for the next 2 items are the same. Select one answer for each item in the set.

For each patient with liver enzyme abnormalities, select the most likely diagnosis.

- ☐ A) Acute cholecystitis

☐ B) Acute pancreatitis

☐ C) Autoimmune chronic active hepatitis

☐ D) Budd-Chiari syndrome

☐ E) Chronic hepatitis B
- ☐ F) Chronic hepatitis C

☐ G) Drug-induced hepatitis

☐ H) Hepatolenticular degeneration (Wilson disease)

☐ I) Primary biliary cirrhosis

☐ J) Sclerosing cholangitis

28. A 36-year-old man comes to the physician because of fever, chills, abdominal pain, and dark urine for 1 day. He was hospitalized for treatment of a similar episode 1 year ago; ultrasonography and CT scan of the abdomen at that time showed no abnormalities. His symptoms resolved after administration of intravenous fluids and antibiotics. He has an 8-month history of pruritus and a 6-year history of ulcerative colitis treated with sulfasalazine. His temperature is 39.3°C (102.7°F), pulse is 108/min, respirations are 16/min, and blood pressure is 125/84 mm Hg. Examination shows jaundice and epigastric tenderness without masses or rebound. The liver is firm and slightly tender and has a span of 12 cm. Laboratory studies show:

Prothrombin time	18 sec (INR=1.8)
Serum	
Protein	
Total	8.0 g/dL
Albumin	4.0 g/dL
Globulin	4.0 g/dL
Total cholesterol	320 mg/dL
Bilirubin, total	3.0 mg/dL
Direct	2.5 mg/dL
Alkaline phosphatase	300 U/L
ALT	120 U/L
HBsAg	negative
Anti-HBs	negative
Anti-HBc	negative
Anti-HCV	negative

For each patient with liver enzyme abnormalities, select the most likely diagnosis.

- ☐ A) Acute cholecystitis

☐ B) Acute pancreatitis

☐ C) Autoimmune chronic active hepatitis

☐ D) Budd-Chiari syndrome

☐ E) Chronic hepatitis B

☐ F) Chronic hepatitis C

☐ G) Drug-induced hepatitis

☐ H) Hepatolenticular degeneration (Wilson disease)

☐ I) Primary biliary cirrhosis

☐ J) Sclerosing cholangitis
29. A 65-year-old man comes to the physician because of easy fatigability for 2 months. In 1985, he had loss of appetite, mild nausea, and fatigue approximately 2 months after receiving a transfusion for a bleeding peptic ulcer. The liver has a span of 12 cm. Examination shows no other abnormalities. Laboratory studies show:

Prothrombin time	12 sec (INR=1.1)
Serum	
Protein	
Total	7.8 g/dL
Albumin	4.5 g/dL
Globulin	3.3 g/dL
Total cholesterol	185 mg/dL
Bilirubin, total	0.8 mg/dL
Direct	0.2 mg/dL
Alkaline phosphatase	70 U/L
ALT	125 U/L
HBsAg	negative
Anti-HBs	positive
Anti-HBc	positive
Anti-HCV	positive

30. A 37-year-old man comes to the physician because of fever and rash for 2 days. He just completed a 6-week course of nafcillin for treatment of staphylococcal bacterial endocarditis. His temperature is 37.8°C (100°F). Examination shows a faint maculopapular rash. Cardiac examination shows an unchanged murmur of aortic insufficiency and no embolic lesions. Laboratory studies and an x-ray of the chest show no abnormalities. Which of the following is the most likely set of findings on urinalysis?

	Specific Gravity	Glucose	Protein	WBC /hpf	RBC /hpf	Casts	Other Findings
☐ A)	1.003	-	1+	30	5	muddy brown	tubular epithelial cells
☐ B)	1.005	trace	2+	30	3	rare WBC	eosinophils
☐ C)	1.012	-	-	2	5	none	-
☐ D)	1.012	-	1+	3	2–3	WBC	2+ bacteria
☐ E)	1.012	-	3+	-	100	RBC	-
☐ F)	1.030	4+	3+	30	30	none	2+ bacteria

31. A 47-year-old man comes to the physician because of a 6-month history of progressive fatigue and shortness of breath on exertion. He has not had difficulty breathing at night or chest pain. In addition to a heart murmur that was diagnosed when he was in high school, he has diverticulosis and mild osteoarthritis. He takes no medications except a daily multivitamin. He is 173 cm (5 ft 8 in) tall and weighs 68 kg (150 lb); BMI is 23 kg/m². His temperature is 37°C (98.6°F), pulse is 82/min, respirations are 18/min, and blood pressure is 135/90 mm Hg. There is no jugular venous distention. The carotid pulse has a slow upstroke. The lungs are clear to auscultation. A grade 2/6 systolic murmur is heard best at the right sternal border and radiates to the apex. There is no change in the murmur with the Valsalva maneuver. There is a paradoxical splitting of S₂ and a prominent apical impulse. An ECG shows left ventricular hypertrophy; there is no evidence of infarction. Which of the following is the most likely underlying cause of this patient's heart murmur?

- ☐ A) Atrial septal defect
- ☐ B) Congenital bicuspid aortic valve
- ☐ C) Hypertrophic obstructive cardiomyopathy
- ☐ D) Myxomatous degeneration of mitral valve
- ☐ E) Rheumatic heart disease





32. A 47-year-old woman is brought to the emergency department because of a 1-day history of temperatures to 39.4°C (103°F) and shortness of breath and a 6-hour history of chest pain exacerbated by cough or inspiration. Three days ago, she started treatment with erythromycin for fever and cough productive of yellow sputum, but she discontinued the medication 24 hours later because of nausea and vomiting. She has autoimmune hepatitis that has been in remission for 3 years. She appears lethargic. Her temperature is 39°C (102.2°F), pulse is 120/min, respirations are 32/min, and blood pressure is 110/70 mm Hg. There is dullness to percussion over the left hemithorax two thirds of the way up from the lung bases. Breath sounds are decreased at the left base. Cardiac examination shows a hyperdynamic precordium with no rubs, gallops, or murmurs. Thoracentesis is performed. Laboratory studies show:

Hematocrit	40%
Leukocyte count	20,000/mm ³
Segmented neutrophils	80%
Bands	15%
Lymphocytes	5%
Pleural fluid	
Leukocyte count	50,000/mm ³
pH	7.11
Glucose	20 mg/dL
Total protein	4 g/dL
Lactate dehydrogenase	400 U/L

Which of the following is the most appropriate next step in management?

- ☐ A) Admission to the coronary care unit
- ☐ B) Chest tube placement
- ☐ C) Instillation of talc in the pleural space
- ☐ D) Intravenous corticosteroid therapy
- ☐ E) Intravenous heparin therapy
- ☐ F) Oral naproxen therapy
- ☐ G) Pericardiocentesis
- ☐ H) Surgical repair

33. A 68-year-old man is brought to the emergency department because of severe substernal chest pain for 2 hours and severe shortness of breath for 30 minutes. He has a history of type 2 diabetes mellitus treated with an oral hypoglycemic agent and hypertension treated with a diuretic. His temperature is 37.5°C (99.5°F), pulse is 104/min, respirations are 28/min, and blood pressure is 100/78 mm Hg. Jugular venous pressure is increased. Bilateral crackles are heard throughout all lung fields. An S₃ and S₄ gallop are heard. Pulse oximetry on room air shows an oxygen saturation of 88%. An x-ray of the chest shows bilateral perihilar infiltrates and a slightly enlarged cardiac silhouette. An ECG shows ST-segment elevation and T-wave inversion in leads V₂ through V₆. Which of the following is the most likely diagnosis?

- ☐ A) Acute pericarditis
- ☐ B) Myocardial infarction
- ☐ C) Pericardial tamponade
- ☐ D) Pneumonia
- ☐ E) Pulmonary embolism



34. An 87-year-old man is admitted to the hospital for treatment of a right femoral neck fracture sustained when he fell down a staircase. He has osteoarthritis of the knees treated with ibuprofen (400 mg three times daily). He takes no other medications. He also reports memory loss, particularly in remembering the names of people he has met recently. He first noted the problem 4 years ago, and he states that it is slowly becoming worse. There is no family history of dementia. He lives alone, drives a car, does his own shopping and cooking, and manages his own finances. He does not smoke cigarettes. He drinks one beer daily. His blood pressure is 138/88 mm Hg. Examination shows shortening and external rotation of the right lower extremity. Neurologic examination shows no focal findings. His Mini-Mental State Examination score is 27/30. Which of the following is the most likely cause of this patient's memory loss?
- ☐ A) Dementia, Alzheimer type
 - ☐ B) Medication adverse effect
 - ☐ C) Mild cognitive impairment
 - ☐ D) Multi-infarct (vascular) dementia
 - ☐ E) Vitamin B₁₂ (cobalamin) deficiency



35. A 64-year-old woman with a 4-year history of type 2 diabetes mellitus controlled with insulin comes to the emergency department because of exertional chest pressure for 3 weeks. The first episode occurred while she was walking upstairs and was relieved by 5 minutes of rest. The second episode occurred 1 week ago while she was mowing the lawn and was relieved after 10 minutes of rest. She had two episodes yesterday, each lasting 15 minutes; both occurred with exertion and were accompanied by shortness of breath and nausea. Today she had a 10-minute episode starting while she was in the shower and another while she was walking into the emergency department. Examination and an ECG show no abnormalities. Which of the following is the most appropriate next step in management?
- ☐ A) Self-monitoring of blood glucose concentration during the next episode of chest pain
 - ☐ B) Dipyridamole-thallium-201 scintigraphy
 - ☐ C) Exercise stress test within 24 hours
 - ☐ D) Antianginal drug therapy now and an exercise stress test in 5 days
 - ☐ E) Admit the patient to the hospital



36. A 48-year-old man with alcoholism is brought to the emergency department confused and drowsy 1 hour after he was found on the ground in a park. Examination shows a lateral gaze defect. He has a wide-based ataxic gait. His blood alcohol concentration is less than 0.10 mg/dL. Administration of which of the following is the most appropriate intervention?

- ☐ A) Folic acid
- ☐ B) Glucose
- ☐ C) Vitamin B₁ (thiamine)
- ☐ D) Vitamin B₆
- ☐ E) Vitamin B₁₂ (cyanocobalamin)



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause

37. A 57-year-old man with a 5-year history of hypertension comes to the physician for a routine examination. He has a 10-year history of throbbing frontal headaches that occur once to twice monthly and are associated with nausea and sometimes with vomiting. Occasionally, he sees halos around objects during an episode. He attributes the headaches to stress at work. His only medication is enalapril. He does not smoke or drink alcohol. He is 178 cm (5 ft 10 in) tall and weighs 100 kg (220 lb); BMI is 32 kg/m². His temperature is 37.2°C (99°F), pulse is 80/min and regular, respirations are 12/min, and blood pressure is 130/85 mm Hg. Examination shows no abnormalities. Serum studies show:

Na ⁺	140 mEq/L
K ⁺	4 mEq/L
Cl ⁻	100 mEq/L
HCO ₃ ⁻	24 mEq/L
Urea nitrogen	18 mg/dL
Glucose	265 mg/dL
Creatinine	1.2 mg/dL
Ketones	negative

Which of the following is the most likely explanation for this patient's increased serum glucose concentration?

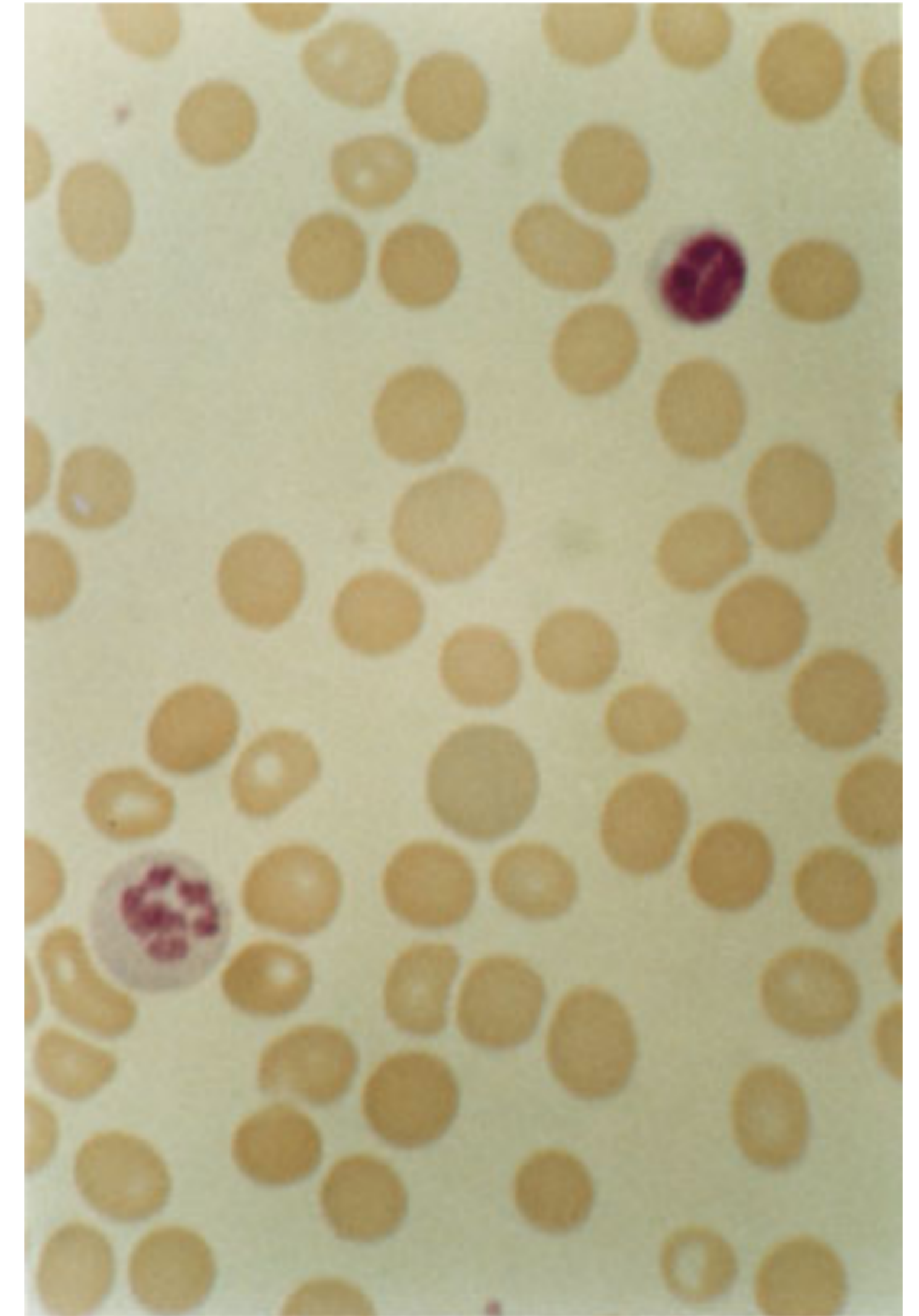
- ☐ A) Cushing syndrome
- ☐ B) Glucagonoma
- ☐ C) Pheochromocytoma
- ☐ D) Somatostatinoma
- ☐ E) Type 2 diabetes mellitus

38. A 27-year-old man is found to have a total serum cholesterol concentration of 190 mg/dL at a routine examination. He has no family history of hypertension or premature coronary artery disease. His grandmother was diagnosed with type 2 diabetes mellitus at the age of 65 years. Examination shows no abnormalities. In addition to general dietary advice and education concerning risk factors, which of the following is the most appropriate next step in management?
- ☐ A) Repeat measurement of serum cholesterol concentration in 5 years
 - ☐ B) Recommend the American Heart Association Step 2 cholesterol-lowering diet
 - ☐ C) Glucose tolerance test
 - ☐ D) X-ray of the chest
 - ☐ E) ECG



39. A 65-year-old woman has fatigue, pallor, and glossitis. Hemoglobin concentration is 8 g/dL, and mean corpuscular volume is $115 \mu\text{m}^3$. Leukocyte and platelet counts are normal. A blood smear is shown. Which of the following is the most likely primary pathophysiologic mechanism causing her anemia?

- ☐ A) Defective porphyrin metabolism
- ☐ B) Faulty protein transcription
- ☐ C) Impaired DNA synthesis
- ☐ D) Inadequate erythropoietin
- ☐ E) Unbalanced synthesis of α - and β -globulin chains



40. A 77-year-old woman comes to the emergency department because of a 3-month history of shortness of breath when she climbs stairs. Her shortness of breath resolves after 5 minutes of rest. Her last episode was 3 days ago. She has not had chest pain, palpitations, orthopnea, cough, wheezing, swelling, or difficulty sleeping. She has hypertension and gastroesophageal reflux disease. Current medications include hydrochlorothiazide, omeprazole, and a multivitamin. She is 168 cm (5 ft 6 in) tall and weighs 63 kg (140 lb); BMI is 23 kg/m². Her temperature is 37.2°C (99°F), pulse is 72/min, respirations are 12/min, and blood pressure is 144/92 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. There are jugular venous pulsations 3 cm above the sternal angle. The remainder of the examination shows no abnormalities. An ECG shows a left bundle branch block and no primary T-wave changes, which is unchanged from an ECG 1 year ago. Which of the following is the most appropriate next step to determine the cause of this patient's dyspnea?
- ☐ A) Measurement of serum troponin I concentration
 - ☐ B) ECG exercise stress test
 - ☐ C) Exercise stress echocardiography
 - ☐ D) Ventilation-perfusion lung scans
 - ☐ E) Coronary angiography



41. Immediately after undergoing an upper gastrointestinal endoscopy and dilatation for achalasia, a 32-year-old woman has moderate substernal and midback pain. She has no history of serious illness. Her temperature is 37°C (98.6°F), pulse is 90/min, respirations are 22/min, and blood pressure is 110/80 mm Hg. The lungs are clear to auscultation. The abdomen is soft and nondistended. Esophagography shows a small leak of contrast from the distal esophagus into the left chest. In addition to intravenous antibiotic therapy, which of the following is the most appropriate next step in management?

- ☐ A) Placement of an intraluminal esophageal stent
- ☐ B) Placement of a left chest tube
- ☐ C) Nissen fundoplication
- ☐ D) Esophagogastrectomy
- ☐ E) Operative repair of the esophageal injury



42. A 33-year-old woman comes to the physician because of a lesion on the thigh for 6 months. The lesion is shown. Which of the following is the most appropriate next step in management?

- ☐ A) Observation
- ☐ B) Radiation therapy
- ☐ C) Laser ablation
- ☐ D) Electrodesiccation and curettage
- ☐ E) Surgical excision



43. A 67-year-old man comes to the physician because of pain of the fingertips for 1 month. Examination shows plethora and splenomegaly. Laboratory studies show:

Hemoglobin	20.2 g/dL
Leukocyte count	14,500/mm ³ with a normal differential
Platelet count	400,000/mm ³

Arterial blood gas analysis on room air shows:

Po ₂	92 mm Hg
O ₂ saturation	94%

Which of the following is the most appropriate next step in management?

- ☐ A) Administration of chlorambucil
- ☐ B) Administration of heparin
- ☐ C) Thrombopheresis
- ☐ D) Phlebotomy
- ☐ E) Splenectomy

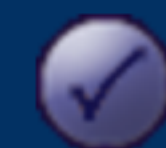
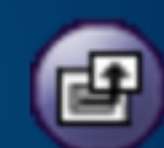
44. A 67-year-old man comes to the physician because of increasing nocturnal incontinence that began 2 months ago and now occurs nightly. He has a 20-year history of hypertension treated with verapamil. He has been taking amitriptyline for major depressive disorder since his wife died 3 months ago. His blood pressure is 158/78 mm Hg. There is lower abdominal distention with diffuse tenderness to palpation. Rectal examination shows a moderately enlarged, smooth prostate with no stool in the vault. Neurologic examinations show no abnormalities. Laboratory studies show:

Serum creatinine	1.6 mg/dL
Urine	
Protein	1+
WBC	1–2/hpf
RBC	2–4/hpf

Which of the following is the most likely diagnosis?

- ☐ A) Benign prostatic hyperplasia
- ☐ B) Cauda equina syndrome
- ☐ C) Dementia, Alzheimer type
- ☐ D) Normal-pressure hydrocephalus
- ☐ E) Urinary tract infection

45. A 52-year-old woman comes to the physician because of a 2-month history of pain and swelling of her knees. She says she also has had stiffness of her knees, which is worse in the morning. During the past month, she has had progressive, generalized fatigue. On examination, the knees are warm, swollen, and tender. Any attempt at passive movement of the knees produces pain. Her hemoglobin concentration is 10 g/dL, leukocyte count is 10,000/mm³, and erythrocyte sedimentation rate is 50 mm/h. X-rays of the knees show osteopenia and subcartilaginous cysts around the joints. The joint spaces are intact. Which of the following is the most appropriate pharmacotherapy?
- ☐ A) Oral alendronate
 - ☐ B) Oral colchicine
 - ☐ C) Oral ibuprofen
 - ☐ D) Intra-articular injection of methylprednisolone
 - ☐ E) Intravenous cefazolin



46. A 32-year-old woman, gravida 1, para 1, comes to the physician because of spontaneous discharge from her nipples during the past 3 months. She discontinued breast-feeding her child 1 year ago. She takes no medications. Menses occur at regular 28-day intervals. On physical examination, a milky white discharge can be expressed from both nipples. Examination shows no other abnormalities. Her fasting serum prolactin concentration is 40 ng/mL, and serum thyroid-stimulating hormone concentration is within the reference range. Which of the following is the most appropriate recommendation for this patient?
- ☐ A) Application of ice packs to the breasts
 - ☐ B) Wearing tight bras
 - ☐ C) Bromocriptine therapy
 - ☐ D) Tamoxifen therapy
 - ☐ E) Vitamin E therapy



47. Four days after undergoing a left total hip replacement, a 75-year-old woman has the rapid onset of chest pain, severe dyspnea, and confusion. Her pulse is 120/min, respirations are 28/min, and blood pressure is 94/60 mm Hg. A pleural rub is heard at the right lung base. The surgical incision appears to be healing well. There is moderate edema of the left leg. Arterial blood gas analysis on room air shows:

pH	7.46
Pco ₂	28 mm Hg
Po ₂	62 mm Hg

Which of the following postoperative treatments is most likely to have prevented this condition?

- ☐ A) Administration of aspirin
- ☐ B) Administration of cefoxitin
- ☐ C) Administration of oxygen
- ☐ D) Administration of warfarin
- ☐ E) Use of incentive spirometry

48. A 57-year-old woman is brought to the emergency department 15 minutes after being involved in a motor vehicle collision. She was the restrained front seat passenger. Her neck was immobilized in a cervical collar at the scene. On arrival, she reports neck pain and pain in her right forearm. Her pulse is 80/min, respirations are 16/min, and blood pressure is 150/70 mm Hg. Examination of the right upper extremity shows swelling and crepitus. She is able to actively extend the proximal and distal interphalangeal joints but not the metacarpophalangeal joints of her right hand or wrist. The remainder of the examination shows no other abnormalities. Plain x-rays show a midshaft fracture of the humerus. Which of the following is the most likely diagnosis?
- ☐ A) Brachial plexus injury
 - ☐ B) Compartment syndrome
 - ☐ C) Herniated cervical disc
 - ☐ D) Median nerve injury
 - ☐ E) Radial nerve injury



49. A 77-year-old woman is brought to the emergency department by her husband because of confusion and sleepiness for the past 12 hours. She has mild hypertension treated with hydrochlorothiazide but is otherwise healthy. Her temperature is 37°C (98.6°F), pulse is 90/min, respirations are 14/min, and blood pressure is 150/100 mm Hg. On examination, she is obtunded and confused. There is no jugular venous distention when she is supine. Cardiopulmonary examination shows no abnormalities. There are no focal neurologic signs. Serum studies show:

Na ⁺	146 mEq/L
K ⁺	5.0 mEq/L
Ca ²⁺	15 mEq/L
Urea nitrogen	40 mg/dL
Glucose	105 mg/dL
Creatinine	1.8 mg/dL
Albumin	3.1 g/dL

Intravenous infusion of which of the following is the most appropriate initial therapy?

- ☐ A) Albumin
- ☐ B) Furosemide
- ☐ C) Glucose and insulin
- ☐ D) 0.9% Saline
- ☐ E) Sodium phosphate

50. A 40-year-old man with alcoholism has had a low-grade fever, malaise, and cough productive of purulent sputum for 2 weeks and hemoptysis for 24 hours; during this period, he has had a 4-kg (9-lb) weight loss. His temperature is 37.8°C (100°F). Hematocrit is 33%, and leukocyte count is 15,000/mm³. An x-ray of the chest shows a large cavitary lesion with an air-fluid level in the superior segment of the right lower lobe of the lung. Which of the following is the most likely diagnosis?

- ☐ A) Aspergilloma
- ☐ B) Bronchiectasis
- ☐ C) Lung abscess
- ☐ D) Lung cancer
- ☐ E) Tuberculosis



Previous



Next

<https://t.me/USMLENBME2CK>

Lab Values



Calculator



Review



Help



Pause